

ASSIGNEMENT N°2 OF CALCULUS FOR ENGINEERS

Group 1.

Question1. Show that the Euler integral of first kind (Beta function)

- a) $B(p, q) = \int_0^1 x^{p-1}(1-x)^{q-1} dx$ converge $p > 0$ and $q > 0$
- b) Calculate $\int_0^1 x^2(a^2 - x^2)^{\frac{1}{2}} dx$

Question2.

- a) Form the sum of the following series $\sum_{n=1}^{\infty} \frac{1+n}{3^n}$ and $\sum_{n=1}^{\infty} \frac{(-1)^{n-n}}{3^n}$ and study the convergence of the results.
- b) Develop $\frac{1}{x^2+4x+7}$ in power series of $x + 2$

Question3.

- a) Consider the matrices $J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

Show that $A^n = 3^n I_3 + a_n J$ where a_n is a real number that you determine.

- b) Find all scalars , if any exist, such that $[1, 0, 1], [2, s, 3], [2, 3, 1]$ are independent?

Question4. Apply the ratio test for studying the convergence of the following series:

$$a) \sum_{k=1}^{+\infty} \left(\frac{3k+2}{2k-1} \right)^k \quad b) \sum_{k=1}^{+\infty} \left(\frac{k}{100} \right)^k \quad c) \sum_{k=1}^{+\infty} \frac{k}{5^k} \quad d) \sum_{k=1}^{+\infty} (1 + e^{-k})^k$$

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Group 2.

Question1.

Find the eigenvalues and the corresponding eigenvectors for the following matrix

$$\begin{pmatrix} 4 & 3 & 1 \\ -4 & -4 & -2 \\ 8 & 12 & 6 \end{pmatrix}$$

Question2 Show that the Euler integral of second kind (Gamma function)

- a) $\Gamma(p) = \int_0^{\infty} x^{p-1} e^{-x} dx$ converge $p > 0$
- b) Deduce the that of $\Gamma(p+1) = p \Gamma(p)$ and of $\Gamma(n+1) = n!$

Question3.

- a) Determine whether the series converges or diverges. If it converges, find the sum:

$$i) \sum_{k=1}^{+\infty} \frac{1}{5^k} \quad ii) \sum_{k=1}^{+\infty} \left(\frac{2}{3}\right)^{k+2} \quad iii) \sum_{k=1}^{+\infty} 4^{k-1} \quad iv) \sum_{k=1}^{+\infty} \left(-\frac{3}{4}\right)^{k-1} \quad v) \sum_{k=1}^{+\infty} (-1)^{k-1} \frac{7}{6^{k-1}}$$

- b) Develop $\frac{1}{x}$ in power series of $x - 1$
- c) Find all scalars , if any exist, such that $[1, 0, 1], [2, s, 3], [1, -s, 0]$ are independent?
- d) Let $A = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$, compute A^2 and A^3 . What will A^{2n} turn out to be?

Question4.

Given $A = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & -2 \\ -6 & 3 \end{bmatrix}$. Solve each of the following matrix equations.

- i) $AX + B = C$ ii) $XA + B = C$ iii) $AX + B = X$ iv) $XA + C = X$

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Group 3.

Question1.

- i) Obtain a reduction formula for the following integral

$$I_n = \int_0^{+\infty} x^n e^{-ax} dx, \text{ where } a > 0$$

- ii) Deduce the value of $\int_0^{+\infty} x^4 e^{2x} dx$

Question2.

- a) Find the surface area generated by revolving the curve $y = \sqrt{1-x^2}$, $0 \leq x \leq \frac{1}{2}$
- b) In electromagnetic theory, the magnetic potential at a point on the axis of a circular coil is given by $u = \frac{2\pi N I r}{k} \int_a^{+\infty} \frac{dx}{(r^2+x^2)^{3/2}}$

Where N, I, r, k , and a are constants. Find u

- c) If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, then calculate A^2 , and $A^n, \forall n \in \mathbb{N} \setminus \{0,1\}$

Question3. Evaluate the following integrals:

- a) $\int \frac{dx}{\sqrt[4]{x} + \sqrt[3]{x}}$
- b) $\int_0^1 \sqrt{2+3x-x^2} dx$

- Question4. a)** Let $A = \begin{pmatrix} \frac{1}{2} & \frac{-1}{2} \\ \frac{-1}{2} & \frac{1}{2} \end{pmatrix}$ Compute A^2 and A^3 . What will A^n turn out to be?

- b) Given $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}$

- i) Compute the determinant of A . Is A nonsingular? ii) Compute $\text{adj } A$ and the product $A \text{adj } A$

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Group 4.

Question1. Show that:

- i) $\mathbf{0}$ is orthogonal to every vector in $\mathbb{R}^2$,
- ii) The vectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 6 \end{pmatrix}$ are orthogonal in $\mathbb{R}^2$,
- iii) The vectors $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ are orthogonal in $\mathbb{R}^3$
- iv) In $C[-\pi, \pi]$ with inner product $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) g(x) dx$ the set $\{1/2, \cos x, \cos 2x, \dots, \cos nx\}$ is orthogonal in $C[-\pi, \pi]$

Question2

- a) Classify the series as: absolutely convergent, conditionally convergent, or divergent

$$i) \sum_{k=1}^{+\infty} \frac{(-1)^{k+1}}{3k} \quad ii) \sum_{k=1}^{+\infty} \frac{\cos k\pi}{k}$$

- b) Use Cramer's rule to solve each of the following systems.

$$i) \begin{cases} x_1 + x_2 = 5 \\ 2x_1 + x_2 - x_3 = 6 \\ 3x_1 - x_2 + 2x_3 = 7 \end{cases} \quad ii) \begin{cases} x_1 + x_2 = 0 \\ x_2 + x_3 - 2x_4 = 1 \\ x_1 + 2x_3 + x_4 = 0 \\ x_1 + x_2 + x_4 = 0 \end{cases}$$

Question3. A box with an open top is to be constructed from a rectangular piece of cardboard with dimensions 14 m by 22 m by cutting out equal squares of side x at each corner and then folding up the sides. Express the volume V of the box as a function of x and find the value of x for which the volume is maximum

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Group 5.

Question1.

- a) Calculate the centre of mass of astroide arc $x^{2/3} + y^{2/3} = a^{2/3}$ which located in first quadrant?

Question2.

- a) Calculate the centre of mass of a figure limited by the curves $y = x^2$, $y = \sqrt{x}$?
b) Classify the series as: absolutely convergent, conditionally convergent, or divergent

$$a) \sum_{k=1}^{+\infty} \frac{\sin k}{k^3}$$

$$b) \sum_{k=1}^{+\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$$

Question3.

- a) Evaluate the following integrals: $\int_0^{+\infty} x^n e^{-ax} dx$, $a > 0$
b) Find the surface area generated by revolving the curve $y = \sqrt[3]{3x}$, $0 \leq y \leq 2$
c) Use the Gauss-Jordan reduction method for solving the following linear system:

$$\begin{cases} x_2 + x_3 + x_4 = 0 \\ 3x_1 + 3x_3 - 4x_4 = 7 \\ x_1 + x_2 + x_3 + 2x_4 = 6 \\ 2x_1 + 3x_2 + x_3 + 3x_4 = 6 \end{cases}$$

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Group 6.

Question1.

- a) Find the eigenvalues and the corresponding eigenvectors for each of the following matrices

i) $\begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix}$ ii) $\begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix}$ iii) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ iv) $\begin{bmatrix} 4 & -5 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$

v) $\begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$

- a) In electromagnetic theory, the magnetic potential at a point on the axis of a circular coil is given by $u = \frac{2\pi N I r}{k} \int_a^{+\infty} \frac{dx}{(r^2 + x^2)^{3/2}}$

Where N, I, r, k , and a are constants. Find u

Question2.

- a) Calculate the total length of curve $r = a \sin^3 \frac{\varphi}{3}$, where $0 \leq \varphi \leq 3\pi$
- b) Find the volume of the solid generated when the region between the graphs of $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ over the interval $[0, 2]$ is revolved about the x-axis?

Question3.

- a) Evaluate the following integrals: $\int_0^{\pi/2} \frac{\cos \theta}{\sqrt{1 - \sin \theta}} d\theta$
- b) Find the surface area generated by revolving the curve $y = \sqrt[3]{3x}$, $0 \leq y \leq 2$

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Group 7.

Question1: Evaluate the integrals that converge: a) $\int_0^8 \frac{dx}{\sqrt[3]{x}}$ b) $\int_{-0}^9 \frac{dx}{\sqrt{9-x}}$ c) $\int_{-2}^2 \frac{dx}{x^2}$
d) $\int_0^{\pi/2} \frac{\cos \theta}{\sqrt{1-\sin \theta}} d\theta$ f) $\int_1^{+\infty} \frac{dx}{x \sqrt{x^2-1}}$

Question2:a) Show that $\int_1^{+\infty} \frac{dx}{x^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

b) Show that $\int_0^1 \frac{dx}{x^p}$ converges if $p < 1$ and diverges if $p \geq 1$.

Question3: a) Given $\vec{x}_1 = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}, \vec{x}_2 = \begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix}, \vec{x} = \begin{pmatrix} 2 \\ 6 \\ 6 \end{pmatrix}, \vec{y} = \begin{pmatrix} -9 \\ -2 \\ 5 \end{pmatrix}$

i) Is $\vec{x} \in \text{Span}(\vec{x}_1, \vec{x}_2)$?

ii) Is $\vec{y} \in \text{Span}(\vec{x}_1, \vec{x}_2)$? Prove your answers.

b) Show that: $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$

Question4: Determine whether the series converges or diverges. If it converges, find the sum:

$$a) \sum_{k=1}^{+\infty} \frac{1}{5^k} \quad b) \sum_{k=1}^{+\infty} \left(\frac{2}{3}\right)^{k+2} \quad c) \sum_{k=1}^{+\infty} 4^{k-1} \quad d) \sum_{k=1}^{+\infty} \left(-\frac{3}{4}\right)^{k-1} \quad e) \sum_{k=1}^{+\infty} (-1)^{k-1} \frac{7}{6^{k-1}}$$

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Group 8.

Question1: Express the repeating decimal as a fraction:

a) $0.4444 \dots$ b) $0.9999 \dots$ c) $0.159159159 \dots$ d) $5.373737 \dots$ e) $0.451141414 \dots$

Question2:a) Let $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}$ If possible, find a matrix B such that $AB = 2A$.

b) Show that the following vectors form a basis for $\mathbb{R}^3$: $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \right\}$

c) Write the vector $\vec{v} = (2, -5, 3)$ in $\mathbb{R}^3$ as a linear combination of the vectors

$\vec{e}_1 = (1, -3, 2)$, $\vec{e}_2 = (2, -4, -1)$ and $\vec{e}_3 = (1, -5, 7)$

Question3: Using geometric series to show:

$$a) \sum_{k=1}^{+\infty} (-1)^k x^k = \frac{1}{1+x} \text{ if } -1 < x < 1 \quad b) \sum_{k=1}^{+\infty} (x-3)^k = \frac{1}{4-x} \text{ if } 2 < x < 4$$

$$c) \sum_{k=1}^{+\infty} (-1)^k x^{2k} = \frac{1}{1+x^2} \text{ if } -1 < x < 1$$

Question4:a) Find the area of the region between the x-axis and the curve $y = \frac{8}{x^2} - 4$ for $x \geq 3$

b) Find the area of the region bounded by $y = 0$, $x = 0$, $x = 4$, and $y = (1-x)^{-2}$

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Group 9.

Question1: Find the sum of series

$$a) \sum_{k=1}^{+\infty} \left(\frac{1}{2^k} + \frac{1}{4^k} \right) \quad b) \sum_{k=1}^{+\infty} \left[\frac{7}{3^k} + \frac{6}{(k+3)(k+4)} \right] \quad c) \sum_{k=1}^{+\infty} \left(\frac{1}{k^2-1} - \frac{7}{10^{k-1}} \right)$$

Question2: Determine whether the given p-series converges or diverges:

$$a) \sum_{k=1}^{+\infty} \frac{1}{k^3} \quad b) \sum_{k=1}^{+\infty} \frac{1}{\sqrt{k}} \quad c) \sum_{k=1}^{+\infty} \frac{1}{\sqrt[4]{k}} \quad d) \sum_{k=1}^{+\infty} \frac{1}{k^\pi} \quad e) \sum_{k=1}^{+\infty} \frac{1}{\sqrt[3]{k^5}}$$

Question3: Show that:

- 0 is orthogonal to every vector in $\mathbb{R}^2$,
- The vectors $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 6 \end{pmatrix}$ are orthogonal in $\mathbb{R}^2$,
- The vectors $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ are orthogonal in $\mathbb{R}^3$
- In $C[-\pi, \pi]$ with inner product $\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) g(x) dx$ the set $\{1/2, \cos x, \cos 2x, \dots, \cos nx\}$ is orthogonal in $C[-\pi, \pi]$

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Group 10.

Question1: Use the divergence test to show that the series diverge:

$$a) \sum_{k=1}^{+\infty} \frac{k+1}{k+2} \quad b) \sum_{k=1}^{+\infty} \ln k \quad c) \sum_{k=1}^{+\infty} \frac{k^2 + k + 3}{2k^2 + 1} \quad d) \sum_{k=1}^{+\infty} \left(1 + \frac{1}{k}\right)^k$$

Question2: Apply the ratio test for studying the convergence of the following series:

$$a) \sum_{k=1}^{+\infty} \left(\frac{3k+2}{2k-1}\right)^k \quad b) \sum_{k=1}^{+\infty} \left(\frac{k}{100}\right)^k \quad c) \sum_{k=1}^{+\infty} \frac{k}{5^k} \quad d) \sum_{k=1}^{+\infty} (1 + e^{-k})^k$$

Question3: Find the eigenvalues and the corresponding eigenvectors for each of the following matrices

$$a) \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} \quad b) \begin{bmatrix} 1 & 1 \\ -2 & 3 \end{bmatrix} \quad c) \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} \quad d) \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad e) \begin{bmatrix} 4 & -5 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$
$$f) \begin{bmatrix} 3 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$